

## Question 1

Write a Python program to store your name, city, and course in variables and display them.

```
name = "Aaryan"
city = "Ahmedabad"
course = "BCA"

print("Name:", name)
print("City:", city)
print("Course:", course)
```

```
Name: Aaryan
City: Ahmedabad
Course: BCA
```

## Question 2

Create variables of type int, float, string, and bool and display their values and datatypes.

```
a = 10
b = 3.14
c = "Hello"
d = True

print(a, type(a))
print(b, type(b))
print(c, type(c))
print(d, type(d))

10 <class 'int'>
3.14 <class 'float'>
Hello <class 'str'>
True <class 'bool'>
```

## Question 3

Write a Python program to demonstrate dynamic typing by assigning an integer value to a variable and then changing it to a string.

```
val = 100
print("Value:", val, "Type:", type(val))

val = "Python Programming"
print("Value:", val, "Type:", type(val))

Value: 100 Type: <class 'int'>
Value: Python Programming Type: <class 'str'>
```

## Question 4

Write a Python program to accept your name and birth year, calculate your age based on the current year, and print your name with the calculated age.

```
name = input("Enter your name: ")
birth_year = int(input("Enter your birth year: "))
current_year = 2026
```

```
age = current_year - birth_year
print("Name:", name)
print("Age:", age)
```

```
Name: aaryan
Age: 28
```

## Question 5

Write a Python program to assign the same value to multiple variables and display them.

```
x = y = z = 50
```

```
print("x =", x)
print("y =", y)
print("z =", z)
```

```
x = 50
y = 50
z = 50
```

## Question 6

Write a Python program to accept length and breadth of a rectangle and calculate its area.

```
length = float(input("Enter length: "))
breadth = float(input("Enter breadth: "))
```

```
area = length * breadth
print("Area of rectangle:", area)
```

```
Area of rectangle: 276.0
```

## Question 7

Write a Python program to accept the radius of a circle and calculate its area.

```
radius = float(input("Enter radius: "))
area = 3.14159 * radius * radius
```

```
print("Area of circle:", area)
```

Area of circle: 6361.719749999999

## Question 8

Write a Python program to perform all arithmetic operations (+, -, \*, /, %, //, \*\*) on two user-entered numbers.

```
a = float(input("Enter first number: "))
b = float(input("Enter second number: "))

print("Addition:", a + b)
print("Subtraction:", a - b)
print("Multiplication:", a * b)
print("Division:", a / b)
print("Modulus:", a % b)
print("Floor Division:", a // b)
print("Exponentiation:", a ** b)
```

Addition: 3.0  
Subtraction: -1.0  
Multiplication: 2.0  
Division: 0.5  
Modulus: 1.0  
Floor Division: 0.0  
Exponentiation: 1.0

## Question 9

Write a Python program to calculate Simple Interest.

```
principal = float(input("Enter Principal Amount: "))
rate = float(input("Enter Rate of Interest: "))
time = float(input("Enter Time (in years): "))

si = (principal * rate * time) / 100
print("Simple Interest:", si)
```

Simple Interest: 3.36

## Question 10

Write a Python program to accept a student's Name, Course, and Semester and display all details in a formatted manner.

```
name = input("Enter Student Name: ")
course = input("Enter Course: ")
```

```

semester = input("Enter Semester: ")

print("-" * 30)
print("Student Details:")
print(f"Name      : {name}")
print(f"Course     : {course}")
print(f"Semester   : {semester}")
print("-" * 30)

```

```

-----
Student Details:
Name      : aaryan
Course     : btech
Semester   : 9
-----

```

## Question 11

Write a Python program to determine whether a student is eligible for examination based on the following conditions: Attendance  $\geq 75$  AND Internal Marks  $\geq 20$  Display the final result (True or False).

```

attendance = float(input("Enter attendance percentage: "))
internal_marks = float(input("Enter internal marks: "))

is_eligible = (attendance >= 75) and (internal_marks >= 20)
print("Eligible for examination:", is_eligible)

Eligible for examination: True

```

## Question 12

Write a Python program to swap two numbers without using a third variable.

```

a = int(input("Enter first number: "))
b = int(input("Enter second number: "))

print("Before swapping: a =", a, ", b =", b)

a, b = b, a

print("After swapping: a =", a, ", b =", b)

Before swapping: a = 87 , b = 78
After swapping: a = 78 , b = 87

```

## Question 13

Write a Python program to check whether a given character exists in a string using the Membership Operator.

```
text = input("Enter a string: ")
char = input("Enter a character to search: ")

if char in text:
    print(f"'{char}' exists in the string.")
else:
    print(f"'{char}' does not exist in the string.")

'C' does not exist in the string.
```

## Question 14

Write a Python program to calculate the square and cube of a number using the exponentiation operator.

```
num = float(input("Enter a number: "))

square = num ** 2
cube = num ** 3

print("Square:", square)
print("Cube:", cube)

Square: 4489.0
Cube: 300763.0
```

## Question 15

Accept a number from the user and classify it as Positive, Negative, or Zero.

```
num = float(input("Enter a number: "))

if num > 0:
    print("Positive")
elif num < 0:
    print("Negative")
else:
    print("Zero")

Positive
```

## Question 16

Accept a number from the user. If it is divisible by 5, print 'Divisible by 5'.

```
num = int(input("Enter a number: "))

if num % 5 == 0:
    print("Divisible by 5")
```

```
else:  
    print("Not divisible by 5")
```

Not divisible by 5

## Question 17

Accept three numbers from the user and print the largest among them.

```
a = float(input("Enter first number: "))  
b = float(input("Enter second number: "))  
c = float(input("Enter third number: "))  
  
if a >= b and a >= c:  
    print("Largest number is:", a)  
elif b >= a and b >= c:  
    print("Largest number is:", b)  
else:  
    print("Largest number is:", c)
```

Largest number is: 90.0

## Question 18

Accept Name, Phone No, age and monthly salary from the user. Using nested if, determine loan eligibility: Age >= 18 AND Salary >= 25000 : Loan Approved Age >= 18 AND Salary < 25000 : Loan Rejected: Low Salary Age < 18 : Not Eligible Age Below 18

```
name = input("Enter Name: ")  
phone = input("Enter Phone No: ")  
age = int(input("Enter Age: "))  
salary = float(input("Enter Monthly Salary: "))  
  
if age >= 18:  
    if salary >= 25000:  
        print("Loan Approved")  
    else:  
        print("Loan Rejected: Low Salary")  
else:  
    print("Not Eligible Age Below 18")
```

Loan Approved

## Question 19

Calculate Profit or Loss by accepting Cost Price and Selling Price from the user.

```
cp = float(input("Enter Cost Price: "))  
sp = float(input("Enter Selling Price: "))
```

```
if sp > cp:
    print("Profit:", sp - cp)
elif cp > sp:
    print("Loss:", cp - sp)
else:
    print("No Profit No Loss")
```

Profit: 11.0

## Question 20

Accept a number N from the user. Using a while loop, print all numbers from 1 to N.

```
n = int(input("Enter N: "))

i = 1
while i <= n:
    print(i, end=" ")
    i += 1
print()
```

1 2 3 4 5 6 7 8 9

## Question 21

Accept a number N. Using a while loop, calculate and print the sum of all numbers from 1 to N.

```
n = int(input("Enter N: "))

total = 0
i = 1
while i <= n:
    total += i
    i += 1

print("Sum:", total)
```

Sum: 36

## Question 22

Accept a number from the user and print its multiplication table from 1 to 10 using a for loop.

```
num = int(input("Enter a number: "))

for i in range(1, 11):
    print(f"{num} x {i} = {num * i}")
```

76 x 1 = 76  
76 x 2 = 152  
76 x 3 = 228  
76 x 4 = 304  
76 x 5 = 380  
76 x 6 = 456  
76 x 7 = 532  
76 x 8 = 608  
76 x 9 = 684  
76 x 10 = 760

## Question 23

Accept a number N. Using a for loop, calculate and print the sum of all even numbers from 1 to N.

```
n = int(input("Enter N: "))  
  
total = 0  
for i in range(1, n + 1):  
    if i % 2 == 0:  
        total += i  
  
print("Sum of even numbers:", total)  
Sum of even numbers: 1980
```

## Question 24

Accept a number N from the user. Using a for loop, calculate and print the factorial of N.

```
n = int(input("Enter N: "))  
  
fact = 1  
for i in range(1, n + 1):  
    fact *= i  
  
print("Factorial:", fact)  
Factorial:  
9426890448883247745626185743057242473809693764078951663494238777294707  
0700232237988829761592077291198236058505886084604294126475673600000000  
00000000000000
```

## Question 25

Accept the number of terms N. Using a while loop, print the Fibonacci series.

```

n = int(input("Enter number of terms: "))

a, b = 0, 1
count = 0

while count < n:
    print(a, end=" ")
    a, b = b, a + b
    count += 1
print()
0 1

```

## Question 26

Accept a number from the user. Using a for loop, check if the number is Prime or Not Prime.

```

num = int(input("Enter a number: "))

if num > 1:
    is_prime = True
    for i in range(2, int(num ** 0.5) + 1):
        if num % i == 0:
            is_prime = False
            break
    if is_prime:
        print("Prime")
    else:
        print("Not Prime")
else:
    print("Not Prime")

Not Prime

```

## Question 27

Accept a year from the user. Check whether it is a Leap Year or a Non-Leap Year.

```

year = int(input("Enter a year: "))

if (year % 400 == 0) or (year % 4 == 0 and year % 100 != 0):
    print("Leap Year")
else:
    print("Non-Leap Year")

Non-Leap Year

```

## Question 28

Accept a number from the user. Find and print all the factors of that number.

```
num = int(input("Enter number: "))

factors = []
for i in range(1, num + 1):
    if num % i == 0:
        factors.append(str(i))

print(f"Factors of {num} are: " + " ".join(factors))
```

Factors of 56 are: 1 2 4 7 8 14 28 56